UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

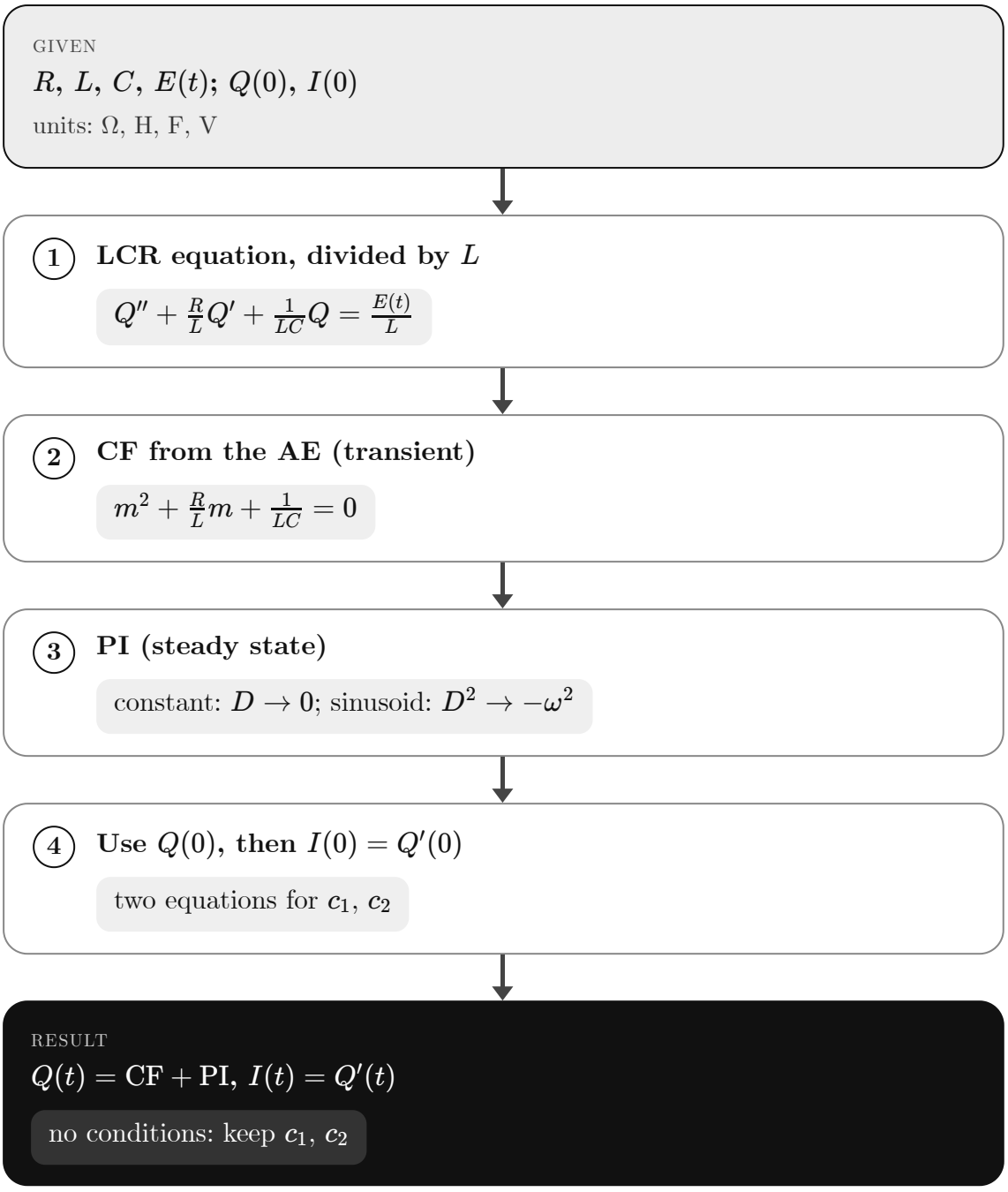
Class 014 · Applications of LDEs: LCR circuits · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S014·8	Unit 2 deck 014 (“Applications of DE – LCR circuit”), slide 8 (rendered .pptx)
T014 00:04:29	Unit 2 AV-summary transcript “Applications of higher-order linear differential equations – RLC Circuits” (11.6 min), segment start
CW U2 P34	Unit 2 class-work problem 34
ESA Jan-24 s3 3b	Past paper: UE23MA141A Jan 2024 set 3, question 3.b

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4. Figure F3.2 · Are initial conditions given?
5. Figure F4.1 · RL circuit, $L\frac{di}{dt} + Ri = E\sin\omega t$
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9. RL, RC and LCR circuits

1 Modelling the series LCR circuit



Remark. Steps 2 and 3 are the class 002 and 003–004 methods.

Exam trap. Divide $E(t)$ by L too.

Figure F1.1. From circuit data to $Q(t)$ and $I(t)$ ESA ★★

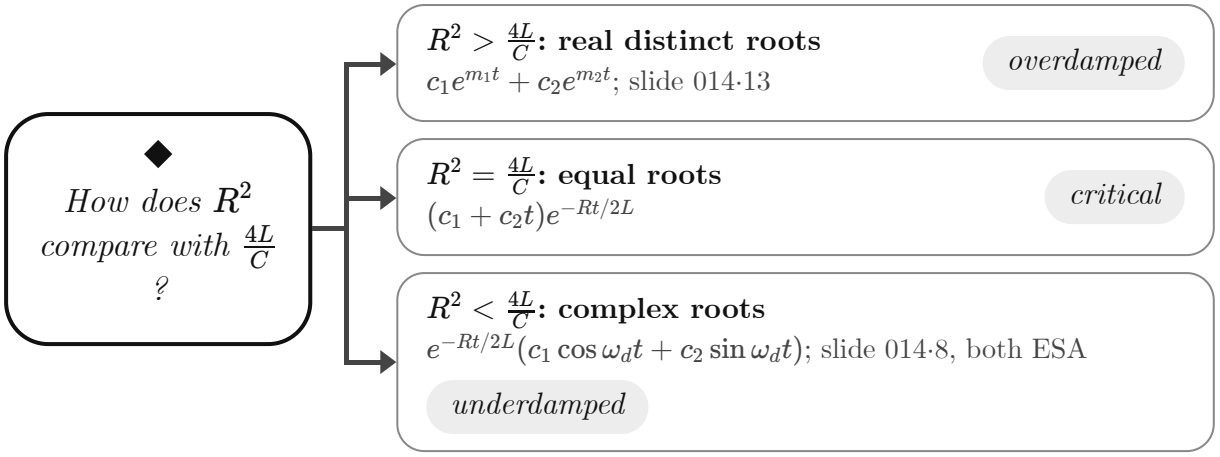
S014·8 · S014·9 · S014·10 · S014·11 · S014·12 · T014 00:04:57 · T014 00:10:31

The three circuit elements

Element	Symbol, unit	Voltage drop	In terms of Q
Inductor	L, henry (H)	$L\frac{dI}{dt}$	$L\frac{d^2Q}{dt^2}$
Resistor	R, ohm (Ω)	RI	$R\frac{dQ}{dt}$
Capacitor	C, farad (F)	$\frac{Q}{C}$	$\frac{Q}{C}$
Source	$E(t)$, volt (V)	supplies $E(t)$	right-hand side

S014·4 · S014·5

2 Solving the circuit equation: CF, PI and damping



Remark. All three decay; only the underdamped one oscillates.

Figure F2.1. Which transient? Compare R^2 with $\frac{4L}{C}$ ESA ★

ADDED FOR CLARITY S014·9 · S014·15

Damping regimes

Test	Roots	Transient	Example
$R^2 > \frac{4L}{C}$	real, distinct, negative	$c_1e^{m_1t} + c_2e^{m_2t}$	slide 014·13: $10^4 > 400$
$R^2 = \frac{4L}{C}$	equal, $-\frac{R}{2L}$	$(c_1 + c_2t)e^{-Rt/2L}$	$R = 2$, $L = C = 1$ (G2.2)
$R^2 < \frac{4L}{C}$	$-\frac{R}{2L} \pm i\omega_d$	$e^{-Rt/2L}(c_1 \cos \omega_d t + c_2 \sin \omega_d t)$	slide 014·8; ESA s3, s4; CW P36

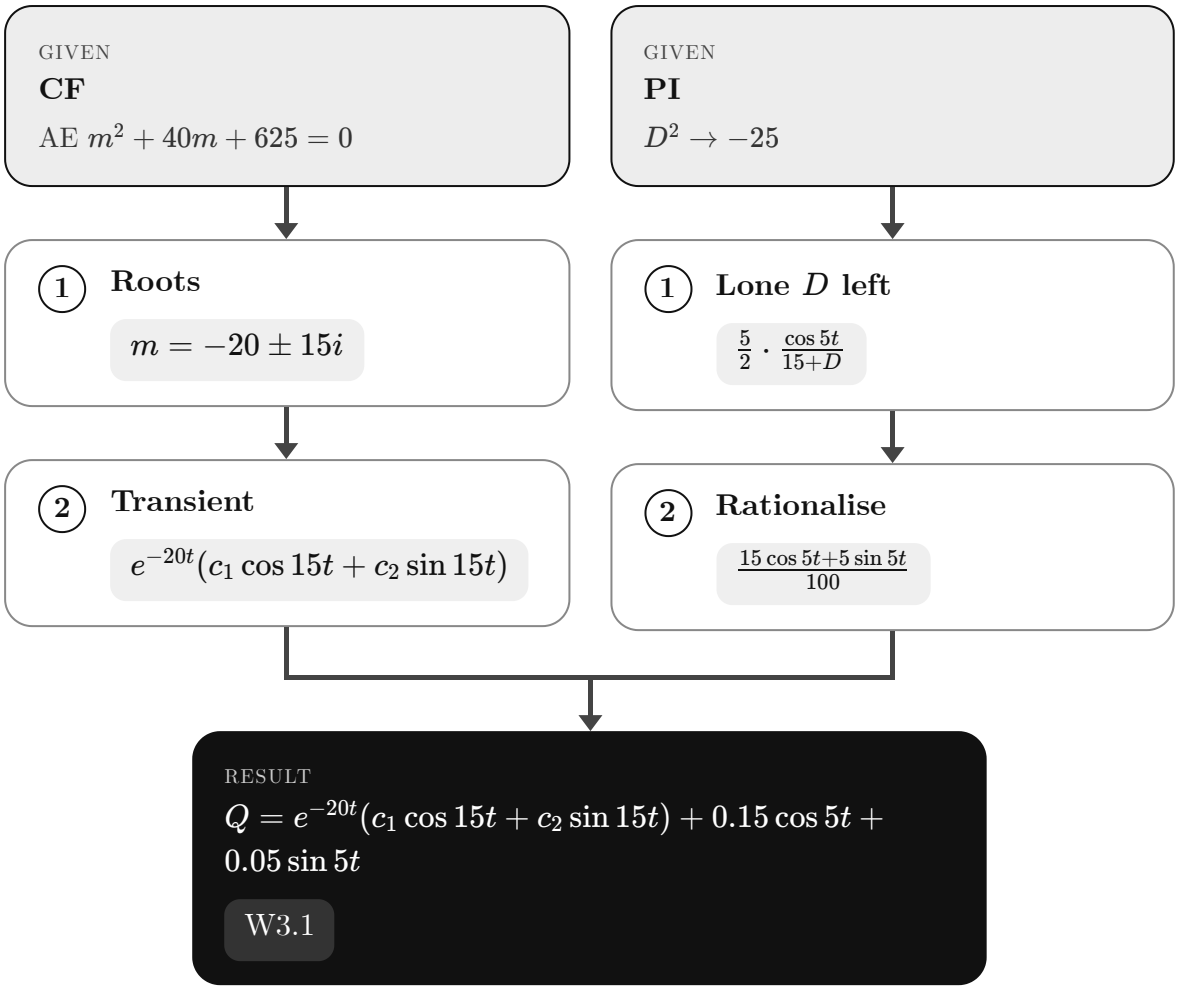
ADDED FOR CLARITY S014·9 · S014·15 · ESA Jan-24 s3 3b

Every circuit problem of this class, side by side

Problem	Normalised equation	Roots	PI (steady state)
Slide 014·8	$Q'' + 4Q' + 5Q = 5$	$-2 \pm i$	1
Slide 014·13	$\ddot{q} + 1000\dot{q} + 10^4 q = 1550 \sin 377t$	$-10.10,\ -989.9$	$-0.00128 \sin 377t - 0.00366 \cos 377t$
ESA s3 3b	$Q'' + 40Q' + 625Q = 100 \cos 5t$	$-20 \pm 15i$	$0.15 \cos 5t + 0.05 \sin 5t$
ESA s4 3b	$Q'' + 20Q' + 500Q = 12 \sin 10t$	$-10 \pm 20i$	$0.024 \sin 10t - 0.012 \cos 10t$
CW P36	$q'' + 1000q' + 2 \times 10^6 q = 0$	$-500 \pm 500\sqrt{7}i$	0 (free)

S014·8 · S014·13 · ESA Jan-24 s3 3b · ESA Jan-24 s4 3b · CW U2 P36

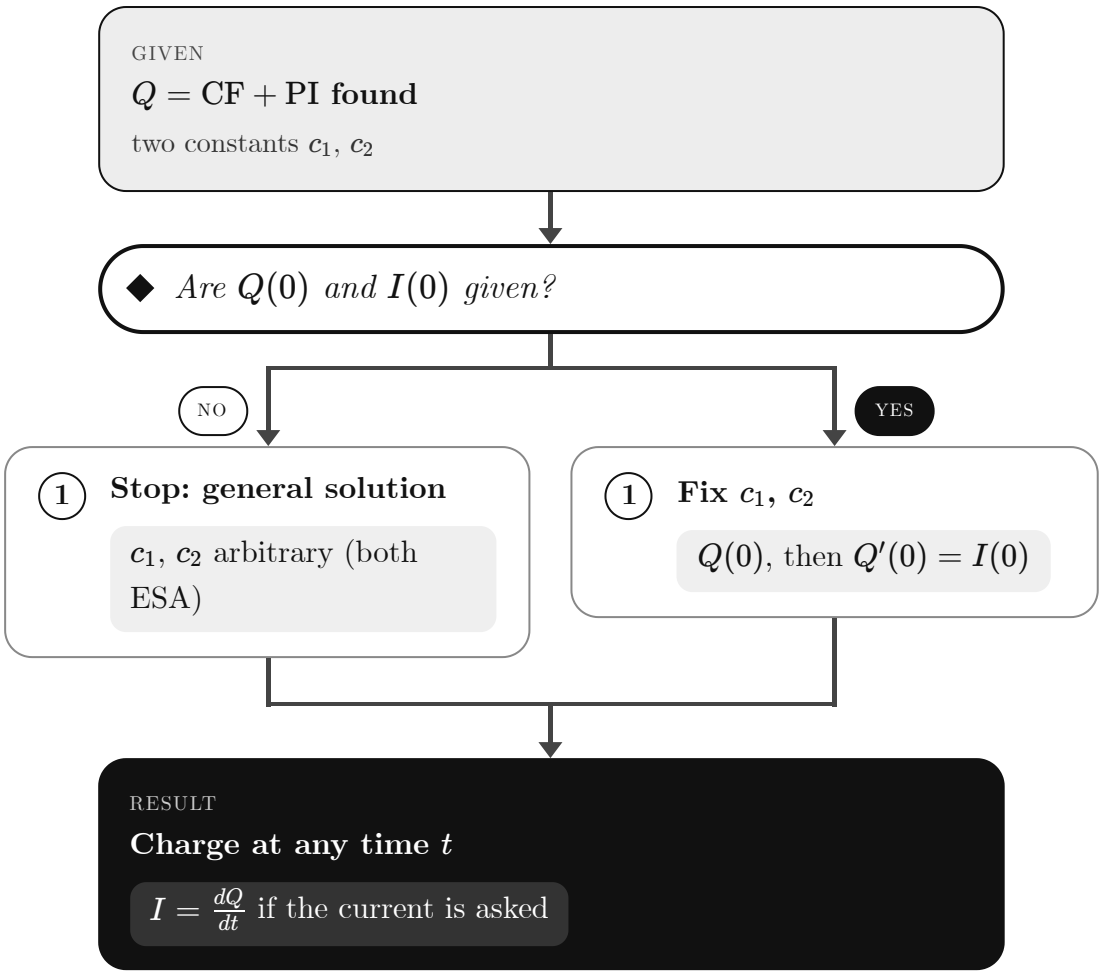
3 Past-paper circuits: AC sources and free discharge



Remark. ESA s4 is the same pattern: $m = -10 \pm 20i$, PI $= 0.024 \sin 10t - 0.012 \cos 10t$ (W3.3).

Figure F3.1. ESA: $Q'' + 40Q' + 625Q = 100 \cos 5t$ ESA ★★

ESA Jan-24 s3 3b

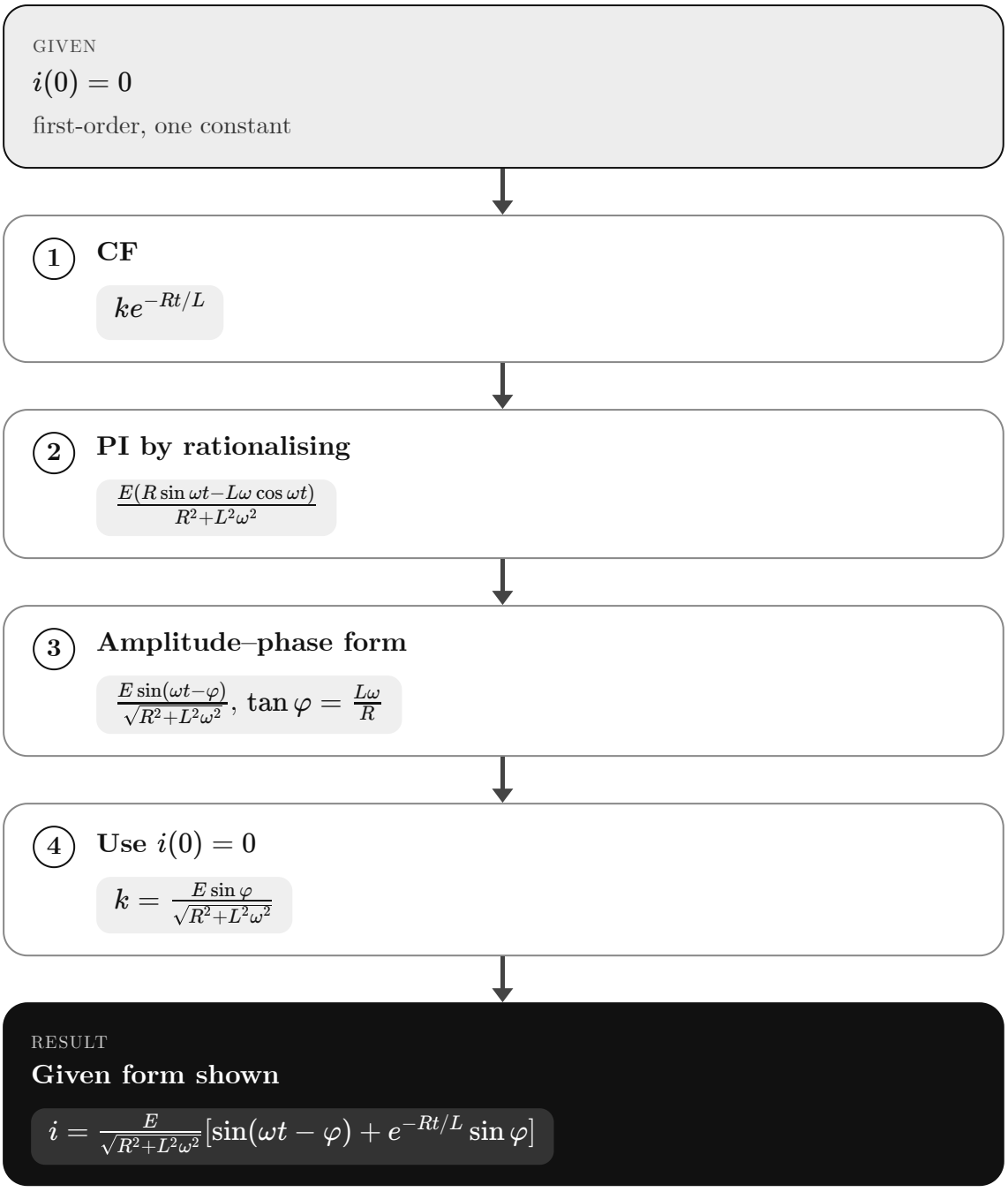


Remark. W3.2 and W3.4 show the "yes" route for both ESA circuits.

Figure F3.2. Are initial conditions given? ESA ★★

ESA Jan-24 s3 3b · ESA Jan-24 s4 3b · S014·11

4 First-order circuits: RL and RC



Remark. The RC circuit (W4.2) follows the same five steps with $m = -\frac{1}{RC}$.

Figure F4.1. RL circuit, $L\frac{di}{dt} + Ri = E\sin\omega t$ ESA ★★

CW U2 P34

RL, RC and LCR circuits

Circuit	Equation	Order	Transient
RL	$L\frac{di}{dt}+Ri=E(t)$	1	$e^{-Rt/L}$
RC	$R\frac{di}{dt}+\frac{i}{C}=\frac{dE}{dt}$	1	$e^{-t/(RC)}$
LCR	$L\frac{d^2Q}{dt^2}+R\frac{dQ}{dt}+\frac{Q}{C}=E(t)$	2	by R^2 vs $\frac{4L}{C}$

CW U2 P34 · CW U2 P35 · S014·6